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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MAR 523 — Past Questions & Materials

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Course: MAR 523

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Paper 1 of 1 — MAR 523 SHIP AUTOMATION SECOND SEMESTER 2023/2024

Level 500 • Second Semester • PastQuestion

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**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
P. M. B. 1221, EFFURUN**

COLLEGE OF TECHNOLOGY

DEPARTMENT OF MARINE ENGINEERING

2ND SEMESTER EXAMINATION, 2023/2024 SESSION

COURSE CODE: MAR 523

COURSE TITLE: SHIP AUTOMATION

INSTRUCTION: ANSWER 5 QUESTIONS ONLY

TIME ALLOWED: 3 HOURS

EXAMINER: E.C. OBINABO, B.Sc. (Portsmouth), M.Eng.(Sheffield), Ph.D.(Nigeria), AMIMEchE.

Q.1. The dynamics of a marine vessel in motion on waves is described by a first order lag. Show that the frequency response locus of the process dynamics is a semi circle located in the first quadrant of the complex frequency plane.

Q.2. The dynamics of an offshore structure for off-loading cargo from a ship is modelled by an exponential lag with unity gain. Automation of the vessel is effected through a direct feedback from the output of the propeller and it takes the speed of the ship 2 seconds to attain 63.2% of its final steady state value.

- (a) Explain what is meant by "automation of the vessel". [5 marks].
- (b) Calculate the speed of the vessel when the delivered power is suddenly changed from 100 to 150 KW.

Q. 3. An offshore marine vessel is powered by a steam reciprocating engine which is connected in cascade with a boiler. Positioning of the vessel on the high seas is accomplished by arranging for the measured position error to manipulate steam generation in the boiler through some suitable amplification. The transfer functions of the steam reciprocating engine and the boiler are:

$$\frac{V_B}{V_E} = \frac{60}{1+0.05s} \quad \text{and} \quad \frac{N}{V_B} = \frac{0.008}{1+0.2s}$$

where V_E denotes amplified position error in the dynamics of the vessel, V_B is the output from the boiler and N is the speed of the vessel.

- (i) Determine the speed of the vessel if the amplified position error is 200 units.
- (ii) Obtain the undamped natural angular frequency in rad/s of the transient response of the vessel for a unity feedback.
- (iii) Explain the effects on the vessel's response of applying the following to the control loop:
 - (a) Derivative of input feedforward.
 - (b) Derivative of error feedforward.

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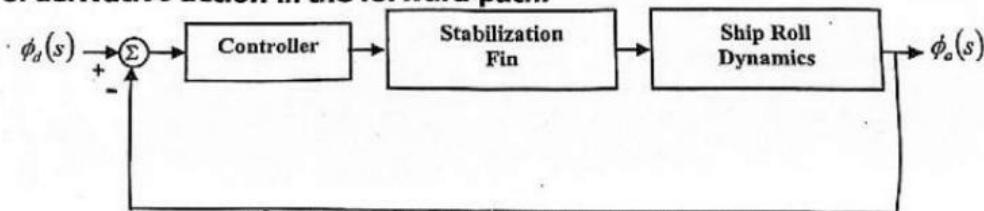
- (iv) If an external load is applied to the vessel, how best can one eliminate the steady state error completely?

Q. 4. (a) Explain briefly what is meant by a root locus with reference to a closed-loop control system.

(b) The forward-path transfer function of a ship roll stabilization system (fig. Q. 4) is given by

$$G(s) = \frac{K'}{s(s+1)(s+6)^2}$$

where K' is adjustable. This system is to be compensated by the addition of derivative action in the forward path.

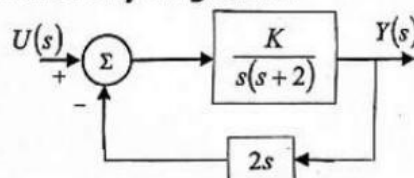


Fig, Q. 4: A Ship Roll Stabilization System

Using the root locus method, determine the values of the proportional and derivative gain necessary for the dominant modes of the system to have a damping ratio of 0.7 and for the system to have a response time of less than 1.5 seconds.

Q. 5. A linear feedback control system has the characteristic polynomial given by the function $s^6 + 2s^5 + 5s^4 + 8s^3 + 8s^2 + 8s + 4$. Determine the even factors of the polynomial, if any, and show that the function has double roots on the imaginary axis of the complex frequency plane. Hence or otherwise, obtain the number of the characteristic roots of the system with positive real part, and explain what this actually means.

Q. 6. For the block diagram of a control system shown below, obtain an associated state space model, and the free response to an input excitation of any magnitude.



Q. 7. Establish a state space model for the process represented schematically below.

